

ASSIGNMENT-10

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BATCH-30

CODE:

```
class Layer2Contract:
```

```
    def __init__(self):
```

```
        self.balances = {}
```

```
        self.gas_fee = 0.0001 # very low
```

```
gas (L2)
```

```
    def deposit(self, address, amount):
```

```
        self.balances[address] =
```

```
self.balances.get(address, 0) + amount
```

```
        print("Transaction Successful")
```

```
        print("Deposited:", amount,
```

```
"ETH")
```

```
        print("Gas Fee (L2):", self.gas_fee,
```

```
"ETH")
```

```
        print("Status: Confirmed
```

```
instantly\n")
```

```
def withdraw(self, address, amount):  
    if self.balances.get(address, 0) <  
amount:  
        print("Transaction Failed -  
Insufficient Balance\n")  
    else:  
        self.balances[address] -=  
amount  
        print("Transaction Successful")  
        print("Withdrawn:", amount,  
"ETH")  
        print("Gas Fee (L2):",  
self.gas_fee, "ETH")  
        print("Status: Confirmed  
instantly\n")
```

```
def show_balance(self, address):  
    print("Wallet:", address)  
    print("Balance:",  
self.balances.get(address, 0), "ETH\n")
```

Execution starts here

```
contract = Layer2Contract()
```

```
user =
```

```
"0x742d35Cc6634C0532925a3b844Bc  
454e4438f44e"
```

```
contract.deposit(user, 10)
```

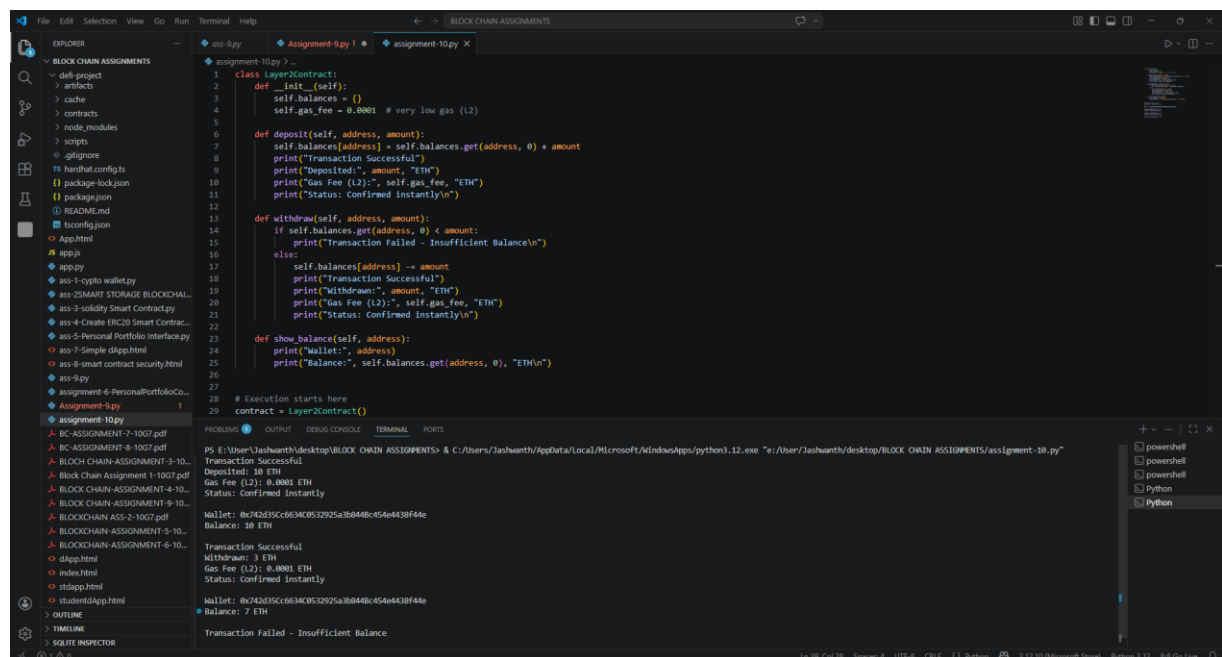
```
contract.show_balance(user)
```

```
contract.withdraw(user, 3)
```

```
contract.show_balance(user)
```

```
contract.withdraw(user, 20)
```

OUTPUT:



The screenshot shows a VS Code editor with a Python script named `assignment-10.py` and its execution output in the terminal. The script defines a `Layer2Contract` class with methods for deposit, withdraw, and show_balance. The terminal output shows the execution of the script, including the creation of the contract, depositing 10 ETH, and withdrawing 3 ETH, resulting in a balance of 7 ETH.

```
1 class Layer2Contract:
2     def __init__(self):
3         self.balances = {}
4         self.gas_fee = 0.0001 # very low gas (L2)
5
6     def deposit(self, address, amount):
7         self.balances[address] = self.balances.get(address, 0) + amount
8         print("Transaction Successful")
9         print("Deposited:", amount, "ETH")
10        print("Gas Fee (L2):", self.gas_fee, "ETH")
11        print("Status: Confirmed Instantly")
12
13    def withdraw(self, address, amount):
14        if self.balances.get(address, 0) < amount:
15            print("Transaction Failed - Insufficient Balance")
16        else:
17            self.balances[address] -= amount
18            print("Transaction Successful")
19            print("Withdrawn:", amount, "ETH")
20            print("Gas Fee (L2):", self.gas_fee, "ETH")
21            print("Status: Confirmed Instantly")
22
23    def show_balance(self, address):
24        print("Wallet:", address)
25        print("Balance:", self.balances.get(address, 0), "ETH")
26
27
28 # Execution starts here
29 contract = Layer2Contract()
30
31 # Deposit 10 ETH
32 contract.deposit(user, 10)
33
34 # Show balance
35 contract.show_balance(user)
36
37 # Withdraw 3 ETH
38 contract.withdraw(user, 3)
39
40 # Show balance
41 contract.show_balance(user)
42
43 # Withdraw 20 ETH
44 contract.withdraw(user, 20)
```

Terminal Output:

```
PS E:\Users\Jashwanth\desktop\BLOCK CHAIN ASSIGNMENTS> python3.12.exe "e:/Users/Jashwanth/desktop/BLOCK CHAIN ASSIGNMENTS/assignment-10.py"
Transaction Successful
Deposited: 10 ETH
Gas Fee (L2): 0.0001 ETH
Status: Confirmed Instantly
Wallet: 0x742d35Cc6634C0532925a3b844Bc454e4438f44e
Balance: 10 ETH
Transaction Successful
Withdrawn: 3 ETH
Gas Fee (L2): 0.0001 ETH
Status: Confirmed Instantly
Wallet: 0x742d35Cc6634C0532925a3b844Bc454e4438f44e
Balance: 7 ETH
Transaction Failed - Insufficient Balance
```

